



SIMATS ENGINEERING

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Course Code: MLA0408

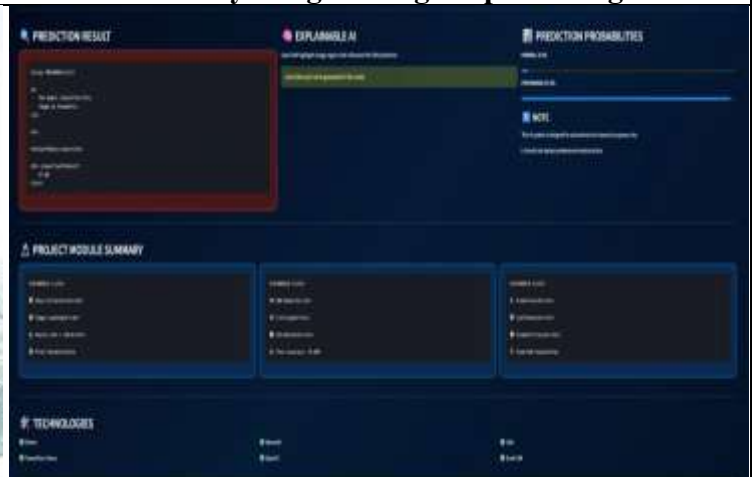
Slot: C

Course Name: Deep Learning

Course Faculty: Dr.D Sudhagar

Project Title:

Intelligent pneumonia detection and diagnosis from chest x-ray images using deep learning.



Project Description: (here you write what you did in this project (contribution) including Model Description)

The Intelligent Pneumonia Detection and Diagnosis from Chest X-Ray Images Using Deep Learning project is designed to develop an automated system for detecting pneumonia from chest X-ray images. Pneumonia is a common respiratory infection that can cause abnormalities in the lungs, and early identification can support timely medical evaluation. The proposed system uses deep learning and image-processing techniques to analyze chest X-ray images and classify them into Normal or Pneumonia categories.

The system begins by collecting and preprocessing chest X-ray images to make them suitable for model training. Preprocessing includes resizing images, normalizing pixel values, and removing unsuitable or incorrect images. Data augmentation techniques such as rotation, shifting, zooming, and flipping can be applied to increase dataset diversity. A Convolutional Neural Network (CNN) or transfer-learning model such as ResNet, DenseNet, or EfficientNet can then learn important visual features such as edges, textures, shapes, and lung abnormalities from the prepared images.

After training, the model can analyze new chest X-ray images and generate a predicted classification as **Normal** or **Pneumonia**, along with a confidence score when available. The system displays the prediction through a simple user interface, making the results easy to understand.

Student Signature

Guide Signature